

Queue Data Structure

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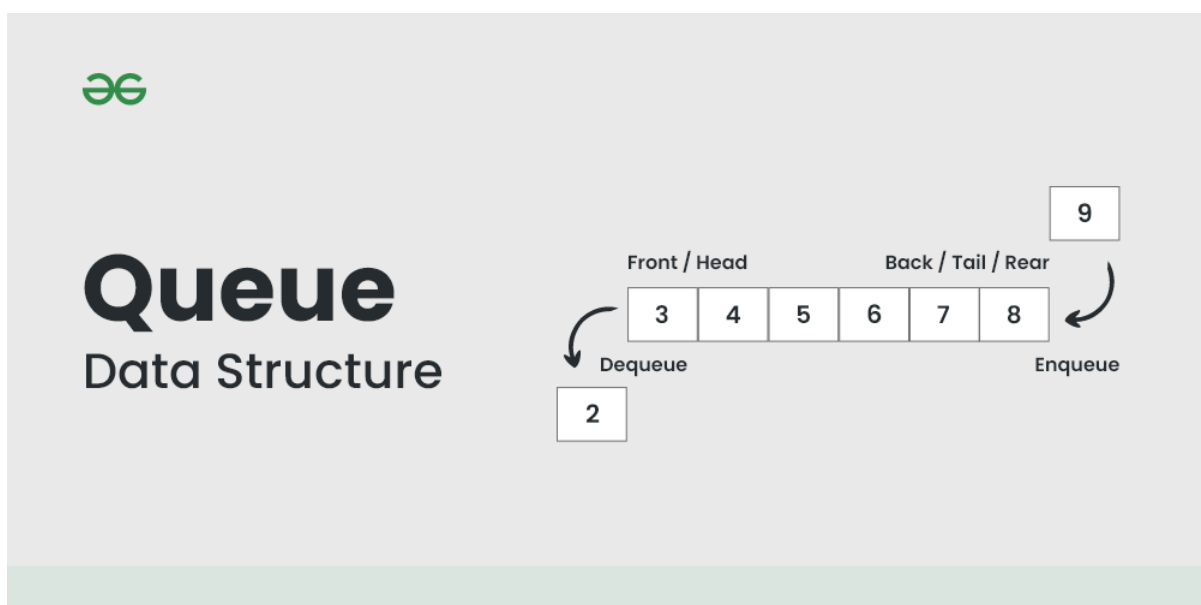


A **Queue Data Structure** is a fundamental concept in computer science used for storing and managing data in a specific order.

- It follows the principle of "**First in, First out**" (**FIFO**), where the first element added to the queue is the first one to be removed.
- It is used as a buffer in computer systems where we have speed mismatch between two devices that communicate with each other. For example, CPU and keyboard and two devices in a network
- Queue is also used in Operating System algorithms like CPU Scheduling and Memory Management, and many standard algorithms like Breadth First Search of Graph, Level Order Traversal of a Tree.

Basics

- [Introduction to Queue](#)
- [Basic Operations of Queue](#)
- [Array Implementations of Queue](#)
- [Linked List Implementation of Queue](#)
- [Applications of Queue](#)



Implementations in various Programming Languages

- [Queue in C++ STL](#)
- [Queue In Java](#)
- [Queue In Python](#)
- [Queue In C#](#)
- [Queue in JavaScript](#)

Implementing Other Data Structures Using Queue

- [Implement a stack using single queue](#)
- [Implement Queue using Stacks](#)
- [LRU Cache Implementation](#)

Easy Problems on Queue

- [Implement Stack using Queues](#)
- [Minimum Depth of a Binary Tree](#)
- [BFS for a Graph](#)
- [Detect cycle in an undirected graph using BFS](#)
- [Right View of a Binary Tree](#)
- [Check whether a graph is Bipartite](#)

Medium Problems on Queue

- [Implement k Queues in a single array](#)
- [Flatten a multilevel linked list](#)
- [Level with maximum number of nodes](#)
- [Find if there is a path between two vertices in a directed graph](#)
- [Print all nodes between two given levels in Binary Tree](#)
- [Find next right node of a given key](#)
- [Minimum steps to reach target by a Knight](#)
- [Islands in a graph using BFS](#)
- [Find the first non-repeating in a stream](#)

Hard Problems on Queue

- [Sliding Window Maximum \(Maximum of all subarrays of size K\)](#)
- [Flood Fill Algorithm](#)

- [Minimum time required to rot all oranges](#)
- [Shortest path in a Binary Maze](#)
- [An Interesting Method to Generate Binary Numbers from 1 to n](#)
- [Maximum cost path from source node to destination](#)
- [Shortest distance between two cells in a matrix or grid](#)
- [Snake and Ladder Problem](#)
- [Find shortest safe route in a path with landmines](#)
- [Count all possible walks from a source to a destination with exactly K edges](#)
- [Minimum Cost of Simple Path between two nodes in a directed and weighted graph](#)
- [Minimum Cost Path in a directed graph via given set of intermediate nodes](#)
- [Find the first circular tour that visits all petrol pumps](#)

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- [‘Quizzes’ on Queue](#)

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Introduction to Queue Data Structure

Queue is a linear data structure that follows FIFO (First In First Out) Principle, so the first element inserted is the first to be popped out. FIFO Principle in Queue:FIFO Principle states that the first element...

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Introduction and Array Implementation of Queue

Similar to Stack, Queue is a linear data structure that follows a particular order in which the operations are performed for storing data. The order is First In First Out (FIFO). One can imagine a queue as a line o...

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Queue - Linked List Implementation

In this article, the Linked List implementation of the queue data structure is discussed and implemented. Print '-1' if the queue is empty.Approach: To solve the problem follow the below idea:we maintain two...

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Different Types of Queues and its Applications

Introduction : A Queue is a linear structure that follows a particular order in which the operations are performed. The order is First In First Out (FIFO). A good example of a queue is any queue of consumers...

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Queue implementation in different languages



Some question related to Queue implementation



Easy problems on Queue



Intermediate problems on Queue

